

**Juan David Muñoz Bolaños**

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com

May 25, 2026

**IMS Nanofabrication GmbH**

Recruiting Team

Dresdner Str. 47, 1200 Vienna, Austria

**Application for the position: Development Engineer**

Dear Recruiting Team,

IMS Nanofabrication drives the limits of lithography at the nanometer scale. As an optical systems engineer finishing my PhD at the Medical University of Innsbruck, I have spent the last four years building complex optomechatronic systems (advanced microscopes) and developing data analysis pipelines to optimize their performance. My background in physics engineering, combined with my exposure to industrial semiconductor workflows at the ASML and ZEISS summer schools, makes me eager to contribute to the development of your Multi Beam Mask Writer systems.

In my current research, I designed and built an automated wavefront sensing and adaptive optics system from scratch. This involved integrating high precision optical components, spatial light modulators, and scientific cameras, while writing the entire control and data acquisition stack in Python and C++. To ensure the system achieved the best potential performance, I developed iterative phase retrieval and optimization algorithms to process wavefront data in real time. This hands-on experience in combining hardware control with advanced signal processing directly aligns with your need for validating new features and analyzing complex log and metrology data.

Beyond system development, I focus on structured testing and quality frameworks. I have structured my instrument development following V-Model systems engineering principles, ensuring clear requirements tracking from initial concept to final validation. By applying Design of Experiments and systematic optical tolerance analysis in python, I have successfully bridged the gap between early stage R&D and stable, reproducible system performance.

I will complete my PhD in June 2026. I am Colombian originally and hold an unrestricted working resident permit in Austria (rot-weiss-rot-plus). I look forward to discussing how my background in optomechanical and electronic systems, data analysis, and structured testing can support your team in Vienna.

Best regards,

**Juan David Munoz Bolanos**

# Juan David Munoz Bolanos

## Junior Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

munozbolanos@gmail.com



### Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

### Research/Professional Experience

Jan 2022 to Jun 2026      **Medical University of Innsbruck**      *Innsbruck, Austria*  
**Optical Systems Engineer / PhD Researcher**

**Optomechatronic System Design:** Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

**Data Analysis & Algorithms:** Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

**Systems Engineering & Quality:** Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

**Lab Management & Safety:** Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019      **INTECOL S.A.S.**      *Medellín, Colombia*  
**Junior Machine Vision Engineer (Field Service)**

**Machine Vision Pipelines:** Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

**Illumination Design:** Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

**Industrial Automation:** Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

### Technical Competencies

#### Software & Tools

Python   C++   LabVIEW   Halcon   OpenCV   ZEMAX   MATLAB   Git

#### Optical Systems

Adaptive Optics   Wavefront Sensing   Microscopy   Spectroscopy   Laser Systems  
DMDs

#### Systems & Quality

V-Model   8D Problem Solving   Design of Experiments   ISO 9001

### Education

Jan 2022 to Jun 2026      **Medical University of Innsbruck**      *Innsbruck, Austria*  
**Doctor of Philosophy in Optical Metrology**

|                      |                                                                                 |                   |
|----------------------|---------------------------------------------------------------------------------|-------------------|
| Oct 2019 to Sep 2021 | <b>Friedrich Schiller Univ. Jena</b><br><b>Master of Science in Photonics</b>   | Jena, Germany     |
| Aug 2012 to Aug 2018 | <b>University of Cauca</b><br><b>Bachelor of Science in Physics Engineering</b> | Popayán, Colombia |

### Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

### Soft Skills

**Adaptability & Independent Working:** Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

**Teamwork & Communication:** Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

### Personal Interests

- Biking/Hiking
- Bachata Dancing
- Harmonica
- Learning German

### Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

### References

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|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Prof. Dr. Monika Ritsch-Marte</b><br><i>Medical University of Innsbruck</i><br>monika.ritsch-marte@i-med.ac.at | <b>PD Dr. Christoph Krafft</b><br><i>Leibniz IPHT</i><br>christoph.krafft@leibniz-ipht.de |
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# **INTECOL**

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

## **TO WHOM IT MAY CONCERN**

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

## **INTECOL**

NIT. 811.041.410-4 - Tel.: 3163822

## **LENY ADRIANA RUIZ PINO**

Administrative and Financial Director

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DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR  
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC  
OF COLOMBIA.

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**UNIVERSIDAD DEL CAUCA**  
**ON BEHALF OF THE**  
**REPUBLIC OF COLOMBIA**  
**AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION**  
**CONSIDERING THAT**

**JUAN DAVID MUÑOZ BOLAÑOS**  
**CC 1.061.772.278 OF POPAYAN**

**HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,**  
**GRANTS THE TITLE OF**

**PHYSICAL ENGINEER**

**WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM**  
**FOR THE PROFESSIONAL EXERCISE**

**POPAYAN, 16 MARCH 2018**

**REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396**  
**RESOLUTION No. R-266-07 MINUTE No. 14-07**

**(SIGNED)**

**THE PRINCIPAL OF THE UNIVERSITY**  
**THE DEAN OF THE FACULTY**  
**THE SECRETARY GENERAL OF THE UNIVERSITY**

*Juan Carlos Ojeda - Garcés*  
TRADUCTOR INTERPRETE OFICIAL  
RESOLUCIÓN No 1171  
MINISTERIO DE JUSTICIA 1997



# URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die  
Physikalisch-Astronomische Fakultät  
mit dieser Urkunde

**Herrn Juan David Munoz Bolanos**  
**geb. 04.06.1994 in La Cruz (Kolumbien)**  
den Hochschulgrad

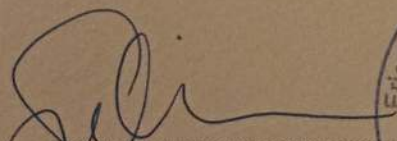
## MASTER OF SCIENCE (M.Sc.)

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Er hat die Masterprüfung (120 ECTS)  
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

  
Prof. Dr. Christian Spielmann  
Dekan



  
Prof. Dr. Silvana Botti  
Vorsitzende des  
Prüfungsausschusses

# Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

## ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

### bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r  
Maria VALLAZZA



ÖSD ist Mitglied von



# ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

# Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

## ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

### bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r  
Maria VALLAZZA



ÖSD ist Mitglied von



# ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

